

Some manufacturers

Resistors

- Cermet Resistors
- Hi-tech Resistors
- Pankaj Potentiometers
- Precision Electronics Component Mfg Co.
- Stead Electronics
- Varitum Electro Products
- Vishay Components
- Watts Electronics

Wound components

- Elcompo Electronic Industries
- GSP Electronics
- Powercaps Systems
- Pulse Magnetic & Power Electronics
- Sinanics Components
- Sree Vishnu Magnetics
- Traffo Electronics
- Victor Components

Capacitors

- Deki Electronics
- Epcos India
- Globe Capacitors
- Gujarat Poly-AXX Electronics
- Incap
- Keltron Electro Ceramics Ltd
- Vishay Components

(Source: ELCOMOS report by ELCINA)

"Although India is increasingly becoming hub for chip design and embedded software, it lags behind in electronics hardware manufacturing capabilities. The electronics ecosystem is quite weak here. There are many challenges to set up manufacturing facilities here, like infrastructure, tax structure, supply chain and logistics, stringent labour laws and limited value addition. As a result, the component distribution has a strong foothold. Imported components are preferred when it comes to critical applications like medical, global EMS or replacements in imported machineries," says Shiv Bhambri, CEO, RS Components & Controls—an electronic component distributor.

"China has large-scale hardware productions for which components

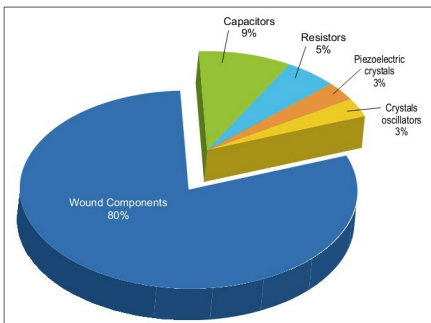


Fig. 2: Passive components' market share (Source: ELCOMOS report by ELCINA)

What are passive components

Passive components, which do not use semiconducting material like germanium or silicon, are largely used to control and provide the energy required for an operation. Resistors, capacitors, piezo-electric crystals and crystal oscillators are some of the widely used passive components.

are needed in big quantities, which is why component producers have established manufacturing presences in China to serve that large market and global demands as well," adds K.N. Krishnamurthy, president-sales & marketing, EPCOS India—a capacitors and cores manufacturer in India.

"However, India is certainly an important market for a number of other application fields, which include power generation and distribution infrastructure, green energy, industrial electronics and, increasingly, household appliances," he feels.

Resistors: Wire-wound dominate

The resistor market in India is largely dominated by the top five-six manufacturers. The overall resistor market size was \$81 million in FY2009-10 with indigenous manufacturing com-

tributing about 27 per cent of the total demand. The market is estimated to have reached a value of \$111 million in FY 2011-12, largely driven by telecommunication, consumer durable and automotive sectors.

Some of the leading resistor manufacturers cater to both the local demand as well as the growing international market. These manufacturers export to countries like Australia, Germany, the UK, Spain, Sweden, Switzerland, Hong Kong, Russia, Middle East and Brazil. The export market was estimated at \$2.83 million in FY 2009-10.

Thin-film and chip resistors currently have relatively low demand in the Indian market. This trend is an indication of the lack of specialised electronic products manufacturing in the country that uses such resistors. However, most manufacturers still feel that the future of resistor manufacturing lies in thick-film and chip resistors.

Wire-wound resistors are the most commonly manufactured resistors in India followed by thin-film resistors. This could be due to the fact that wire-wound resistor manufacturing plants do not need significant capital investments and are labour-intensive. In case of other types of resistors, the demand